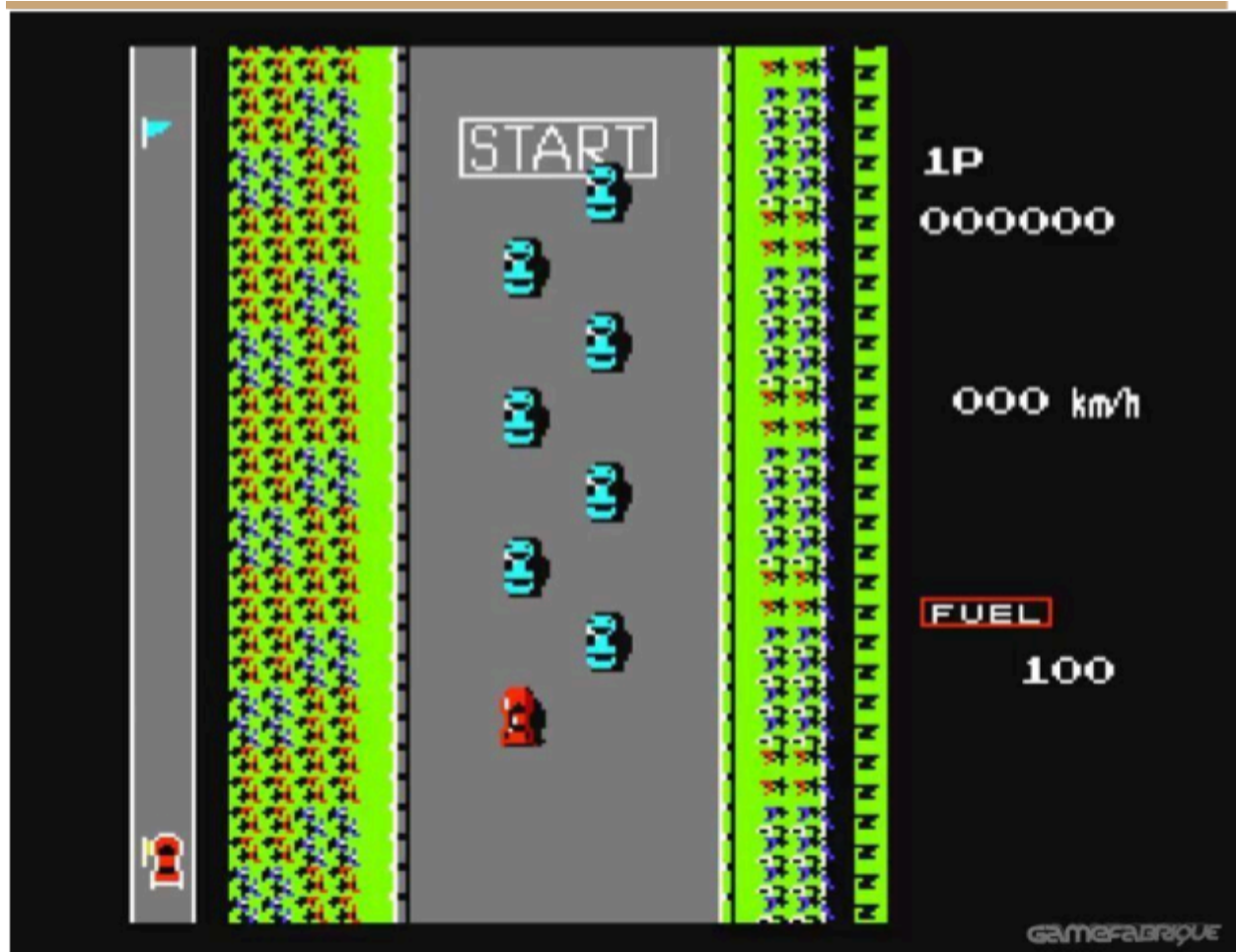


COL215

Laboratory Assignment Report 8 (Part 3)

Road Fighter Game



Created by:

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Lab Assignment

In Part III, pseudo-random number generation and collision detection logic extend the Road Fighter game to include a dynamically spawned rival car with autonomous vertical movement. An 8-bit LFSR generates randomized horizontal positions for the rival car, which descends at a controlled frame-based speed and respawns upon reaching the road boundary. Collision detection identifies hit-box overlaps between the main car and rival sprite, triggering game state transitions.

The system comprises of:

- **Modules:**
LFSR-based random position generator, rival_car display controller, frame counter for velocity regulation, and collision detection logic.
- **Random Generation & Movement:**
The rival car spawns at random horizontal positions within road boundaries and descends vertically at configurable frame intervals. Upon reaching the bottom, it automatically respawns at a new random position.
- **Collision Detection:**
Rectangular hit-box overlap detection identifies collisions between the main car and rival car sprites, transitioning the FSM to a COLLIDE state.
- **Output Display:**
VGA output renders both cars with proper sprite layering and real-time collision feedback.
- **Deliverables:**
Verilog code for LFSR, rival car control, and collision detection.
Comprehensive testbenches for random generation, movement timing, and collision cases.
Validated .xdc constraint and bitstream files for Basys3.
Report (4–5 pages) with simulation results, FSM diagrams and utilization data.

Design Explanation and implementation

LFSR-Based Random Number Generation:

- 8-bit Galois LFSR with feedback polynomial (XOR taps at) generates pseudo-random sequences
- Initializes to `LFSR_CONST = 8'b01001111` (derived from group kerberos IDs) on system reset
- Shifts and feeds back on each clock cycle at 3, 4, 5 and 7 index,, producing new 8-bit random value per cycle
- Random value maps to horizontal rival spawn position: `rival_x = OFFSET_BG_X + 44 + (random_num mod 61)`

Moving Background Logic:

The background implements vertical scrolling by maintaining a `bg_y_offset` register that increases at each frame refresh (excluding the COLLIDE state). For each pixel on screen, the displayed background pixel is selected using:

$$\text{rom_addr} = (\text{hor_pix} - \text{OFFSET_BG_X}) + ((\text{ver_pix} - \text{OFFSET_BG_Y} + \text{bg_y_offset}) \% 160) \times 240$$
$$\text{rom_addr} = (\text{hor_pix} - \text{OFFSET_BG_X}) + ((\text{ver_pix} - \text{OFFSET_BG_Y} + \text{bg_y_offset}) \% 160) \times 240$$

This causes the background image to seamlessly scroll upward, wrapping around for a continuous loop and giving the effect of road motion behind the cars.

Rival Car Sprite and Vertical Movement:

- 14×16 pixel sprite stored in dedicated ROM (initialized from .coe file)
- Renders using region-based address computation: `ROM_address = (hor_pix - rival_x) + (ver_pix - rival_y) × 14`
- Pink transparency (12'b1010_0000_1010): pink pixels show background; non-pink show sprite
- Frame-counted descent: `frame_count` increments at video frame boundaries (`hor_pix = 799, ver_pix = 479`)

- After every FRAMES_BUFFER frames (default 3), `rival_y` increments by `RIVAL_Y = 2 + random_num * 2` pixels
- Spawns at `y = OFFSET_BG_Y - 16` (top); respawns after reaching `y = OFFSET_BG_Y + 240` (bottom)

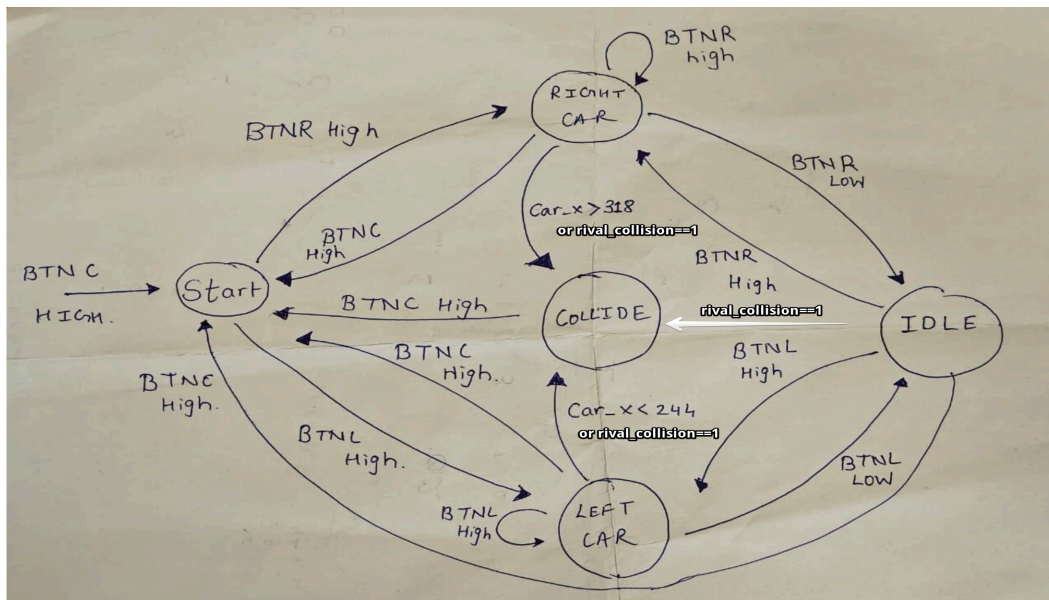
Rival Car Collision Detection (AABB Overlap):

- Uses axis-aligned bounding box (AABB) method for rectangular overlap
- Collision detected when ALL four conditions hold simultaneously:
 - `car_x < rival_x + 14` (main car's right edge past rival's left edge)
 - `car_x + 14 > rival_x` (main car's left edge before rival's right edge)
 - `car_y < rival_y + 16` (main car's bottom edge past rival's top edge)
 - `car_y + 16 > rival_y` (main car's top edge before rival's bottom edge)
- Upon collision, `collision_detected` flag asserts high; FSM immediately prioritizes COLLIDE transition

Debouncer Module:

- Eliminates mechanical switch jitter using per-bit threshold counters
- For each input bit: when input differs from registered output for THRESHOLD consecutive cycles (default 3), output updates
- Prevents spurious FSM state changes and ensures stable button transitions

FSM Details:



1. START State (0)

- Purpose: Initializes game (car and rival positions, random LFSR seed).
- Transition:
 - Moves to IDLE if no button is pressed after reset.
 - Moves to LEFT or RIGHT if player presses btnL or btnR (debounced).

2. LEFT State (1)

- Purpose: Main car moves left when btnL is pressed.
- Transition:
 - Stays in LEFT as long as btnL is held (debounced), no collision occurs.
 - Moves to RIGHT if btnR is pressed (debounced), overriding left.
 - Moves to IDLE if neither btnL nor btnR is pressed.
 - Jumps immediately to COLLIDE if collision is detected at any cycle (collision_detected = 1 or by edge collision).

3. RIGHT State (2)

- Purpose: Main car moves right when btnR is pressed.
- Transition:
 - Stays in RIGHT as long as btnR is held (debounced), no collision occurs.
 - Moves to LEFT if btnL is pressed (debounced), overriding right.
 - Moves to IDLE if neither btnL nor btnR is pressed.
 - Jumps to COLLIDE if collision_detected = 1 or edge collision detected at any point.

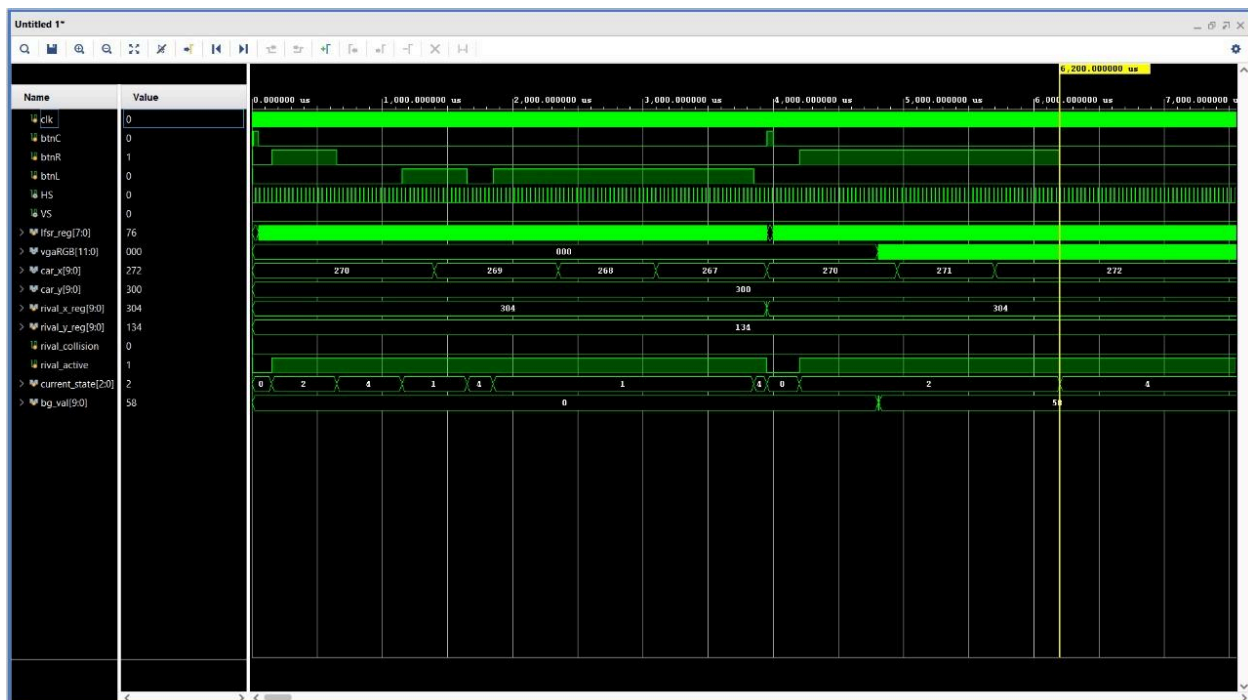
4. IDLE State (4)

- Purpose: No player input; car stays still, rival continues descending, background scrolls.
- Transition:
 - Moves to LEFT if btnL is pressed.
 - Moves to RIGHT if btnR is pressed.
 - Remains in IDLE if no button is pressed and no collision.
 - Directly transitions to COLLIDE if collision detected (either with rival car or edge collision).

5. COLLIDE State (3)

- Purpose: Signals game over; car and rival freeze.
- Transition:
 - Remains in COLLIDE as long as reset (btnCdebounced) is not pressed.
 - Returns to START on reset (btnCdebounced), reinitializing all positions.

Simulation Result



The simulation waveform shown here represents mainly:

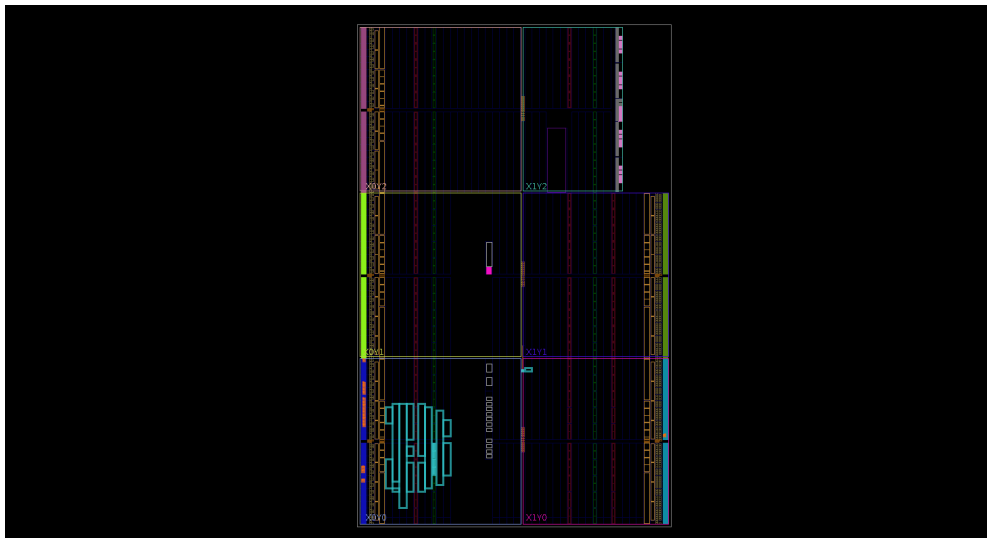
- The left and right movement of the car by the discrete updation of the car_x variable which increase/decrease by 1 pixel if a counter reaches multiple of specific value.
- The rival car is active as the variable "rival_active" is high.
- The "rival_on" variable is LOW throughout as in the timeframe shown here, the Hcnt and Vcnt couldn't reach the the pixel where the display actually begins.
- **"State Transitions"**: with the buttons control, the specific state transition can be observed trivially. Start, Left, Right, Collide, Idle are 0,1,2,3,4 respectively.



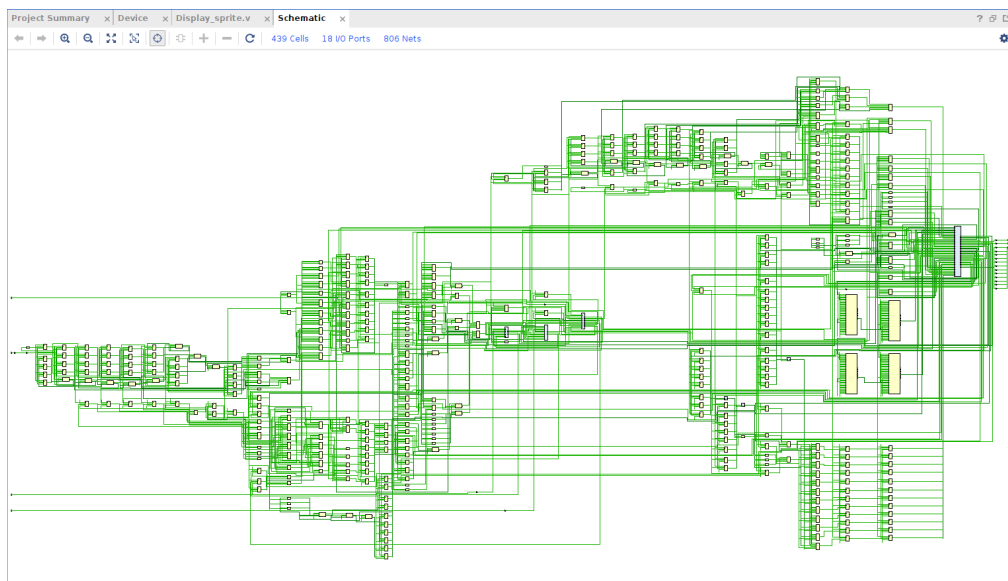
This waveform is here to represent:

- Random number generation for the rival_x_reg value 304, 362 and 259 here. Assignment of new random number is prevented when the collision is detected.
- Steady increase in rival_y_reg (with a value of 30 here (kept this high only for simulation purposes)) till it reaches the bottom of the screen, after which it restarts from the top with new x value.
- The rival car collision detected as variable "**rival_collision**" gets HIGH with change of state to collision almost instantly. The related transition is from 4 (IDLE state) to 3 (COLLISION state).
- The display of rival car is activated once the game starts.
- The variable bg_val, which is responsible for the movement of background by logic already explained, stopped increasing as long as the car is hit.

Design suite



Schematic Diagram (of combined gates)



Synthesis Report

Tcl Console																		Messages																		Log																		Reports																		Design Runs																		x 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Tcl Console				Messages	Log	Reports	Design Runs	x DRC	Methodology	Power	Timing			
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				11/4/25, 2:11 PM	00:00:16	Vivado Synthesis Defaults (Vivado Synthesis 2022)	Vivado Synthesis Default Reports (Vivado Synthesis 2022)	xc7a35tcpg236-1	dhd	Vivado Synthes				
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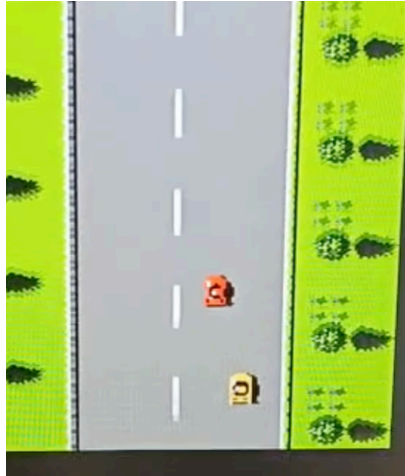
1. LUTs=888
2. Flip flops=251
3. BRAM=0
4. DSPs=4

Programming the Basys 3™ FPGA Board

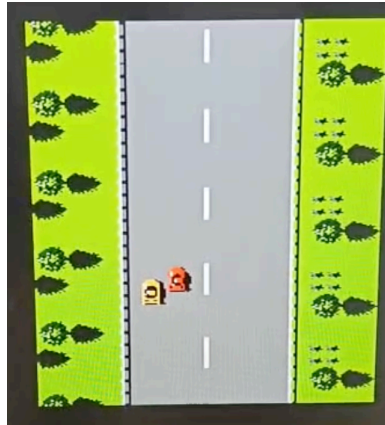
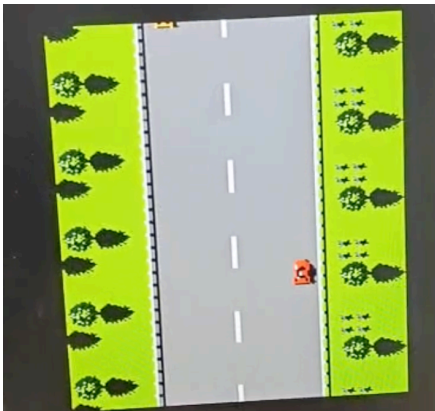
After generating the bitstream of the system, using the constraint files and source file:

- **Constraint file** containing the data of mapping and active switches/7-segment displays.
- **Source file** containing the logic according to which the board will be programmed.
- **The .bit file** is programmed inside the Basys 3™ FPGA Board.

Here is the demonstrated functioning of board:



Normal driving of the car (without collision)



Car colliding with the the rival car and the edge